

Assessing and managing important risks

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A common problem in project risk management processes is the need to determine the relative significance of different sources of risk so as to guide subsequent risk management effort and ensure it remains cost effective.

A common approach is to rank risks in terms of probability and impact to identify sources of risk which will receive the most attention. This paper examines the shortcomings of this technique in guiding the analysis and management of risks and considers the information needed for a proper assessment of importance. For cost-effective management it is desirable to distinguish not only between the size of impacts and probability of impacts occurring, but also other factors such as the nature of feasible responses, and the time available for responses. The paper offers some practical suggestions for dealing with this problem. © 1999 Elsevier Science Ltd and IPMA. All rights reserved

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For many busy managers with problems on all sides, perhaps one of the hardest tasks is deciding which problems to work on first. In a Harvard Business Review article entitled “How senior managers think”, Isenberg quotes one manager’s description of this difficulty:¹

“I have to sort through so many issues at once. There are ten times too many. I use a number of defence mechanisms to deal with this overload—I use delaying actions, I deny the existence of problems, or I put problems in a mental queue of sorts. This is an uncomfortable process for me”

The project risk manager’s position is very similar. For some risks procrastination may be a sensible option, and denying the existence of some risks may be occasionally expedient, particularly if the consequences are likely to be borne by other parties to the project. However, faced with a large number of sources of risk, and the impracticality of attempting to manage them all, the natural tendency is to seek to put identified risks in some kind of order of priority. This may be an ‘uncomfortable process’ to the extent that it is difficult to identify *a priori* which risks are the most significant, and how many risks are worth identifying and managing as specific, individual items. Any techniques which offer help with this prioritising task, especially simple ones, are likely to find willing users.

A related consideration is that with typically limited time and resources, any effort expended on analysis and management of project risks needs to be cost-effective. Thus risk assessment and management may be biased towards relatively simple procedures, per-

haps with some notion that 80% of the effect can be achieved with 20% of the effort. Simple procedures also have the advantage of being less demanding in terms of information requirements and are perhaps more readily understood.

Probably the most common approach to risk assessment and management involves the use of a summary risk register which lists project risks and associated information in a tabular format. More detailed information about each individual risk may be documented separately to allow more detailed consideration of each risk as necessary. The purpose of the summary risk register is to help the project team review project risks on a regular basis throughout the project. A tabular form of risk register has the attraction of simplicity and convenience, but has a number of shortcomings, particularly if unsupported by other backup documentation. These shortcomings include:²

1. individual risk drivers may not be described in sufficient detail to avoid ambiguity and misunderstandings about what risk is being described;
2. important inter-dependencies between risks are not readily highlighted;
3. a table of risk drivers, particularly a long one, provides limited guidance on the relative importance of individual risk drivers.

These three shortcomings are linked in that shortcoming 1 contributes to shortcoming 2, and shortcoming 2 in turn, contributes to shortcoming 3.

Most tabular risk registers include a summary evaluation of each risk to indicate its relative size. This

		Probability		
		Low	Medium	High
Impact	Low	1	1	2
	Medium	1	2	3
	High	2	3	3

Figure 1 'Qualitative' scoring in a probability-impact grid

assessment is used to rank risks for management purposes.

A very common approach to ranking risks involves the use of probability-impact grids. Typically such grids require individual sources of risk to be characterised as risk events with a roughly assessed probability of occurrence and degree of impact. This allows each risk to be characterised by a single 'risk rating'. The higher the risk rating the more important the risk, in some sense. Highly rated risks can expect to receive more attention from risk managers than lowly rated risks. However, what form this extra attention should take is not entirely clear. Certainly different considerations apply in the different phases of the risk management process, as discussed later.

Limitations of probability-impact grids

Figure 1 illustrates a commonly used form of probability-impact grid. Probabilities and impacts of individual risks are assessed qualitatively as 'High' (H), 'Medium' (M) or 'Low' (L). For illustrative purposes, cells of the grid are labelled 3, 2 and 1 to indicate classes of risk considered to be of the highest, intermediate and lowest importance respectively. Variations might use other qualitative labels, more categories, or more classes of risk.

Sometimes numerical scales are used to 'score' each risk in terms of impact and probability of occurrence and thereby generate a rank ordering of risks.³ Figure 2 gives a very simple example of such a grid. With this format, each cell has an associated risk rating which is calculated as the product of the two scores:

$$\text{Risk rating} = (\text{Impact score}) \times (\text{Probability score})$$

Clearly, risk ratings calculated in this way have no absolute meaning. For example, in Figure 2 it would be inappropriate to conclude that a risk rated 100 is more than one hundred times 'more important' than a

risk rated 1, or twice as important as a risk rated 50 etc. It follows that adding together the ratings for individual risks to indicate an overall amount of risk exposure on a project is of little value, and any temptation to do so should be resisted.

Calculating risk ratings in this way is appealing because the formula resembles that for unconditional expected impact = (expected impact $\times$ probability). One way to order sources of risk would be to place them in descending order of expected impact. However even ranking risks according to expected impact is unsatisfactory as the process ignores potentially relevant information about the spread of possible impacts.⁴

Quantifying the categories used in a probability-impact grid makes classification of risks less subjective, but further highlights the shortcomings of ranking cells by multiplying probability and impact scores associated with each category together for each cell. Figure 3 illustrates a grid which uses four well defined ranges for impact in terms of expected cost (given that the risk occurs), and three well defined ranges for the probability that the impact occurs. Each cell shows the implied range of unconditional expected cost for each pair of impact and probability categories. It is immediately obvious that the ranges in different cells overlap substantially. This will be the case in general, although the precise extent of overlaps will depend on the ranges chosen to categorise probability and impact.

How then should the grid cells in Figure 3 be ranked? Where cells have associated expected cost ranges which are similar then it might be argued that such cells should be equally ranked: thus cells (2,1) and (3,1) might be equally ranked as more important than the equally ranked cells (1,1) and (1,2). Ranking the other cells given the overlapping expected cost ranges is more problematic, and the value of doing so must seem questionable.

Rating risks is a policy decision

The forgoing section demonstrates that there are major conceptual problems in justifying a ranking of risks by calculating risk ratings as a product of numbers relating to 'expected impact' and 'probability of occurrence'. Accordingly, attributing different levels of importance to risks corresponding to different cells in a probability-impact grid may be best treated as a policy decision which is not necessarily described by a convenient formula or decision rule.

		Probability		
		Low - score 1	Medium - score 5	High - score 10
Impact	Low - score 1	1	5	10
	Medium - score 5	5	25	50
	High - score 10	10	50	100

Figure 2 'Numerical' scoring in a probability-impact grid

		Probability, p		
		p < 0.1	0.1 < p < 0.2	0.2 < p < 1.0
Impact, i (£k)	0 < i < 10	(0,1)	(0,2)	(0,10)
	10 < i < 100	(0,10)	(1,20)	(2.5,100)
	100 < i < 500	(0,50)	(10,100)	(20,500)
	500 < i	0 <	50 <	100 <

Figure 3 A twelve-way classification of cost risks

In principle each cell could have a unique label, thereby indicating which categories of impact and probability a given risk is associated with. If time and resources are limited, a relatively informal approach may be expedient, where risk managers use their judgment and experience in the context of a particular project to rate the significance of individual risks to the extent that they find this personally useful. To underline this approach, users of probability-impact grids might find it more helpful to label individual cells alphabetically rather than in numerical terms, with ranking of cells reflected by alphabetical order.

The use of an alphabetic scale could highlight the subjective nature of assessments and remove the temptation to employ quantified ratings in spurious ranking calculations which may convey a misleading impression of precision and objectivity.

Multiple risk dimensions

The above discussion used cost impact to illustrate the arguments, but impacts may also be in terms of time or some aspect of quality. Some approaches to ranking risks try to incorporate information about cost, time and quality impacts for a given risks by using a separate probability-impact grid for each type of impact. Often such approaches use 'numerical scoring' (as in Figure 2) for each impact and then add the three separate ratings together to give a single composite rating for each risk. However, as previously noted, the separate ratings only represent ordinal rating. Adding ratings for cost, time and quality together obscures information and implies trade-offs between cost, time and quality of an unspecified nature, even assuming such trade-offs are feasible. Applying different weights to the ratings for cost, time and quality before adding them together merely compounds the problem of implicit trade-offs.

Where a multi-dimensional assessment of risks is required, a more transparent approach would be to apply alphabetic ratings based on the use of these grids, one for each type of impact. Explicit consideration of the relative significance of different combinations of the three dimensional ratings would then be required.

The attraction of the probability-impact grid approach is that neither 'probability of occurrence' or 'conditional expected impact' for a given risk have to be specified very precisely. Thus the approach is particularly convenient in the early stages of a risk analysis when the analyst may have limited information about individual risks, and is trying to target further

analysis. The danger is that their use may lead to the analysis of risk being confined prematurely, and exclusively, to initially identified 'high probability', 'high impact' risks.

Important risks in phases of the risk management process

Whatever their limitations, the popularity of probability-impact grids and associated risk rating is a reflection of the felt need to indicate important risks over less important ones, and perhaps, to do this relatively quickly and simply. Typically the intention is that more highly rated risks should receive closer attention from (designated) project personnel. However, what this 'closer attention' should consist of is unclear, whether in terms of more detailed analysis, or in terms of subsequent management of responses.

To explore this question of important risks further, it is necessary to consider the tasks the risk manager needs carry out. The Association for Project Management's project risk analysis and management (PRAM) Guide describes these tasks in terms of a six phase process, summarised below:³

- Define the project;
- Focus the PRAM effort;
- Identification of risks, and possible responses;
- Assessment involving structuring and ownership of risks and responses, estimation of uncertainty and evaluation of implications;
- Planning recommended proactive and reactive contingency plans;
- Management of planned responses, monitoring and controlling developments and revising plans as necessary.

The question of which risks are important arises in the latter four phases, and in iterations through these four phases at intervals throughout the project life cycle.

Important risks in the Identification phase

It is often said that the real risks in any project are the ones that you fail to identify. This suggests a broadly scoped effort to identify potential sources of risk and thereby reduce the chances of overlooking important areas of risk. The emphasis should therefore be on generating a comprehensive list of risks rather than seeking prematurely to identify a limited set of key risks.

During the identification of risks there is a natural tendency to simply omit recording some risks because their impacts are immediately considered to be of a minor nature. This has obvious dangers in that:

1. apparently minor problems which do not have an effective response may be overlooked;
2. the combined effect of large numbers of apparently minor risks may be underestimated.

When a risk is first identified, at least one response must be implicit in order to describe the impact of an identified risk, the default response being 'do nothing and accept the risk' (which may not be feasible). A preliminary consideration of responses can offer some clues to the relative importance of identified risks. For example, some large impact risks under a 'do nothing' scenario may have very effective responses which can be readily and reliably implemented. Such risks are less problematic than risks for which only partially effective responses are identified or for which significant response formulation effort and investment may be required.

Important risks in the Assessment and Planning phases

In the Assessment phase, an appreciation of which risks are important can help managers to decide:

1. whether to explore particular risks in more or less detail;
2. how much time to invest in developing responses to particular risks; and
3. how much resource it is worth investing in responses to particular risks.

The Assessment phase should begin with attempts to structure the data about risks and responses from the previous Identification phase. The objective of this is to improve understanding of the relative importance of different sources of risk given identified responses, to explore more thoroughly the interactions between project activities, risk and responses, and to test the assumptions implicit in all earlier steps. This can lead to a refinement of existing responses and prompt the development of new, more effective responses. Failure to recognise important structural features of project risk can cause analysis to be dangerously misleading. For example, if the adverse impacts of two or more sources of risk are positively correlated (things tend to go badly at the same time), such circumstances need to be appreciated. 'Small' impact risks should not necessarily be dismissed as unimportant.

Failure to look for inter-dependencies between risks can also lead to missed opportunities. For example, some responses to particular risks may serve as general responses in that they can deal with whole sets of risks, possibly all risks up to that point in a project, including risks which have not been identified. It is important to recognise the opportunities provided by such general responses, and the value of investing resources in such responses.

Following efforts to explore interactions between project activities, risks and responses, it is necessary to assess the relative importance of different sources of risk in order to guide subsequent risk analysis and management effort. Chapman and Ward² suggest that it can be useful at this point to formally identify some

risks as 'minor', to eliminate them further from consideration (if their impact is negligible), or to collect them together into a 'risks treated collectively' category. Hopefully this leaves a relatively small residual of 'risks to be treated separately' which are all 'major' risks by definition. This distinction can help to determine the level of effort to be expended on further analysis of particular risks and more detailed development of associated response plans.

As a general principle the larger the potential impact and the higher its probability of occurrence, the more resources should be expended on responses to manage the associated risk, and the more time and effort it is worth expending on identifying and developing effective responses. For PRAM resource allocation purposes, an assessment of importance based on a probability-impact grid may be useful in a first iteration through the Identification and Assessment phases. However, more refined comparison of impacts associated with different risks may be needed if responses look as if they will require trade-offs to be made in the subsequent, simultaneous management of different risks with limited resources.

Important risks in the Management phase

The Management phase is primarily concerned with monitoring changes in risk exposure and implementing planned responses. It is the characteristics of planned responses that influences the intensity with which particular risks are actively managed and monitored.

In the Management phase the importance of risks needs to be assessed on a time dimension as well as an impact dimension. Risks may still be more or less important in the sense employed in the Identification and Assessment phases, but now effective use of management time requires that risks and their associated responses also take on a priority ranking reflecting trends in each risk, and the urgency of associated responses.

The question of urgency concerns the level of time pressure behind a given risk and its associated responses. Specific concerns here are how soon action must be taken, how long it will take to identify and implement a course of action, and how much time there is to make a decision.

At any point in time, some very high impact risks may require less urgent attention than some lower impact risks. Some risks may arise at any time and recur through the life of the project. Such risks require continuous attention unless effective action to remove them can be taken. Some risks may be more likely to occur in the short term, in an imminent stage of the project works. Other risks may be unlikely to occur until a later stage in the project, with the possibility that action to manage the risk in the immediate future can take a more passive form, for the time being.

Urgency may be mitigated by the scope for interim, stop-gap action which might gain time for further analysis and postpone costly deployment of resources. Of course, urgency is reduced if risks have been identified at an early stage. There is then more time to identify, assess, plan for and implement responses, and usually the range of possible responses is wider and with more effective and efficient responses available. In a strategic context, this means being responsive to

what Ansoff⁵ calls 'weak signals', when risks may be ill-defined and decisive responses difficult to formulate. In these circumstances, responses will be correspondingly unfocused, preferably emphasising flexibility and robust courses of action, with refinement of responses as information about a risk becomes more precise. What may have initially looked like a small, unimportant risk, may develop on its own or interact with other factors to threaten much larger impacts than was first recognised. This underlines the danger of relying on a single, initial identification of 'important risks', with or without the employment of probability-impact grids to rank risks.

An important aspect of prioritising risks in the Management phase involves monitoring trends in identified risks and progress in the implementation of responses. The relative importance of risks inevitably changes as a project progresses for a variety of reasons, and these changes need to be tracked. For example, as a project progresses and the project plan is reviewed, risks which have been realised or avoided can be eliminated from further consideration, and confidence band assessments of project performance should contract, unless new risks are envisaged. Plotting how this process is progressing can be useful, especially if some serious set-backs have been experienced, but the chance of achieving commitments is stable or improving.

Plotting perceptions of exposure to persisting risks can also be very useful. For example, will a particular risk persist at previously estimated levels if no action is taken, or will it get worse? If a particular risk does look as if it is getting to be more of threat, at what point will new responses need to be implemented? Additionally, there may be new risks which flare up and need managing along side of existing response plans.

Conclusion

The desire to identify key risks in a project is a natural one, based on a recognition that resources available to manage risks are limited, and that risk management efforts need to be cost-effective. To achieve this, the information included in risk registers ought to include information about the timing of risks and responses, resources required by alternative responses, and information about inter-dependencies, as well as other basic information such as the nature of impacts and responses, and risk ownership. Consequently, *Table 1* offers a list of items that might be included in an individual risk register.

The contents of *Table 1* imply a fairly detailed document for each risk, from which the relative importance of risks is to be determined from time to time as the associated project progresses.

To facilitate comparisons between risks, common practice is to extract summary data from each individual risk register and summarise this in tabular form on a summary risk register. Inevitably it is this summarised data which is focused on in determining the relative importance of individual risks. The result is often a simplistic comparison linked with the use of some form of probability-impact grid. Notwithstanding their low information requirement, probability-impact grids need to be used with care, recognising the substantial

Table 1 Items for a risk register

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- risk identifier, title and description
 - description of causes and trigger events
 - description of impacts on cost, time and quality and quantitative assessment of range of impacts where appropriate
 - nature of any inter-dependencies with other sources of risk
 - timing of likely impacts
 - probability of occurrence
 - description of feasible responses, including timing required
 - resource implications of responses
 - likely effect of responses on this risk
 - nature of any significant inter-dependencies with other risks and responses
 - residual risk after effective response
 - party bearing the consequences of the risk
 - party responsible for managing the risk and implementing responses
-

limitations for risk management purposes of merely ranking risks in order of ratings based on probability and impact scores.

The key message of this paper is that the importance of risks typically rests on other factors besides probability and impact, and different considerations can apply in different phases of the risk management process. In the Identification, Assessment and Planning phases of the PRAM process, appropriate attention ought to be paid to the range of feasible responses, and the nature of inter-dependencies between risks and responses. Probability-impact grids may have a role in guiding an initial 'first cut' at response development, but the real importance of risks can only be assessed after further iterative analysis which explores structural linkages and responses in more detail.

Subsequently, when ongoing implementation of responses and monitoring of risks is the concern, risk management effort needs to be concerned with priorities in developing and implementing responses, rather than the relative size of associated risks. The imminence of particular risks and the time needed to put in place an appropriate response need to be appreciated alongside of assessments of probability and impact.

Summary risk registers are undoubtedly a useful management tool, but warrant more careful design if they are to be the main basis for determining priorities in the risk management process. The forgoing discussion suggests that consideration of rather more than a simple characterisation of impact and probability of occurrence are required. Accordingly, summary registers might usefully include other pertinent data about each risk, to reflect concerns at various phases of the PRAM process.

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